Project

Visualisation for IA (MIA) 2025/26

Introduction

The main goal of the information visualisation project is the development and application of a set of skills that allow the construction of interactive visualisations capable of solving problems related to Artificial Intelligence models. The work is organised in a manner simulating the conditions that occur in a real-world project. This includes activities such as communication with the client (teacher) and obtaining feedback, working with data, identification of the problem(s) and possible solutions, sketching and prototyping, implementation of interactive visualisation, and validation. It is also intended that the students demonstrate skills in analysis and synthesis, organisation and planning, critical reasoning, application of knowledge in practice, investigation, autonomous learning, and group work.

The project must integrate the following characteristics and requirements:

1. All the code should be produced by the students;
2. Students are free to utilise Generative AI tools in an ethical manner;
3. Students are free to use any visualisation tools and frameworks;
4. Students should use Plotly/Dash framework and Python to implement interactive visualisations;
5. Students can use any other framework compatible with the Plotly/Dash frameworks (e.g., Vega-Altair, D3.js, etc.);
6. Students are free to use any additional frameworks to support the main code;
7. The data should be aligned with the theme of the project and be approved by the teachers;
8. The visualisation should be an analytical application;
9. The visualisation should implement at least one advanced visualisation technique (e.g., bar graphs, line charts, and pie charts are not advanced techniques);
10. Include other interactive elements, such as hover, range selection, or any other covered during the semester;
11. The visualisation should follow good practices of design (i.e., contain all the necessary graphical elements, such as legends, titles, etc.).

Groups

The project should be developed in groups of **2 elements**. The groups should be defined by filling out this form: VIA Groups 2025/26

Milestones

The project is divided into several milestones with respective deadlines:

- **Group formation** and initial brainstorm (first week): We expect you to form a team as soon as possible and to start brainstorming your ideas for the project.
- **Project submission** (10th of January): For the final submission you should hand in the following items (see details in Project section): i) report and ii) the code.
- **Project presentation and defences** (last day): There will be a final defence of your projects. You should start with a formal presentation of your project, after which the defence continues with a Q&A session. Defences are mandatory. The projects will be evaluated according to predefined grading criteria (see respective Section). The defences require registration for a time slot, which will be opened in the *Inforestudante* closer to the defence date.

Evaluation

The grading for the project is equal to 80% of the final grade, which is equivalent to **16 out of 20.**
Important notice! It is not possible to improve the grade of the project.
Another important notice! Submissions made after the deadline will not be accepted.

Grading criteria

The following elements will be evaluated:

- Solution:
    - Application of visualisation principles and practices of good design (Was it designed following the principles studied in the course?);
    - Visual interaction (Does your visualisation employ interaction methods?);
    - Effectiveness and appropriateness of the solution (Is your visualisation effective in answering the defined questions? Is your visualisation effective in communicating data/insights? Does your solution employ appropriate visualisation methods and encodings?);
    - Advanced visualisation (Does your solution include advanced visualisation models? Does your application include advanced interaction techniques?).

- Implementation: Quality of the implementation (Is your visualisation robust? Is your application optimised to run in real-time? Does your code follow conventions and overall good programming practices?)
- Presentation and defence: Clear, engaging, and effective presentation. Did you answer all the questions during the defence? The score for this component is individual.
- Written report: Are your project and the progress well documented in the report?

Minimum grade

To get approval in the Visualisation for AI course, it is necessary to obtain a classification equal to, at least, 40% of the final grade of the project.

Project

The project consists of creating a visualisation to help users through an AI model lifecycle. The students are faced with real-world datasets with the goal of identifying AI related problems and developing a visualisation that should solve these problems. The visualisation should be effective, should contain multiple views, and should incorporate appropriate visualisation techniques for the defined theme of the project. The students should choose one of the following themes: i) data preparation; ii) model architecture and training; iii) model explainability and interpretability; iv) (optional) model evaluation. Finally, the visualisation should follow good practices of visual design, as well as the principles and theories studied in this course. Search for inspiration in the class slides or design studios (e.g., https://www.nand.io/, https://www.pitchinteractive.com/, https://truth-and-beauty.net/).

Project themes

The practical project should focus on one of the following themes, which correspond to different phases in an AI model lifecycle:

- **Data preparation**: this is the first phase in any AI related project. The focus should be on different aspects of data preparation, which will later be used for model training. So, the students can choose to work on the following:
    - data cleaning;
    - data exploration and analysis (create visualisations for exploratory data analysis EDA);
    - data fairness and bias (create visualisations to analyse bias in a dataset);
    - feature selection (create visualization help with feature selection and engineering);
    - other characteristics (create a visualisation to help explore other characteristics of a dataset).
- **Visualisation of model architecture and training process**: the focus should be on helping the users in the creation of a model:

    - parameters fine-tuning (create a visualisation to explore the effect on parameters);
    - Architecture understanding (create visualisations to facilitate the analysis of model architecture);
    - training (create a visualisation to monitor the training process);
    - other identified by the students.
- **Model explainability and interpretability**: the focus should be on opening the black box of AI model. The students can choose work on the following:
    - decision-making (create visualisations to help understand why and how a model makes decisions locally and globally);
    - feature attribution and importance (create visualisations to depict the weight of inputs);
    - bias audition (create a visualisation to understand if the model relies on proxy attributes);
    - trust (create visualisations to gain trust of a general audience);
    - others identified by the students.
- **Model performance evaluation**:
    - focus on traditional performance metrics, such as accuracy (precision and recall, ROC curve, etc.);
    - faithfulness (create visualisations to validate how explanations match the internal model's logic);
    - fairness and bias (create a visualisation to audit performance across different demographic groups. Is the model fair for everyone?);
    - robustness (create a visualisation to validate consistency of the model);
    - other proposed by the students.

Progress report (the article)

Keeping track of your progress is an important part of your project. Remember, it is equally important how you get your final results! Therefore, each project should be documented in the progress report, providing details about each part of your project, including any additional information that you find relevant (e.g., alternative designs you tried, the problems that you faced, or the insights that you learned). It is also important to underline that your report book is not a logbook or diary. You should present your work in an informative and concise way, without unneeded details that do not contribute to the understanding of your project. Instead, include many figures in your progress report (e.g., different designs you tried, photos of your sketches, or sketches you made using different visualisation tools, etc.). Ultimately, the goal is to convey the progress and rationale of your project in a clear and informative way.

The process report should not exceed 8 pages in length (plus references). We suggest including 3-4 pages of text and 5-4 pages of figures. Figures should be placed near their reference in the text.

Data

Students are free to use any data and datasource. We strongly encourage you to explore interesting data visualisation problems. Students are allowed to use any additional data sources to enrich the datasets at hand. Try to be creative and employ all the skills you have earned in the other courses. Additionally, you can explore public repositories for the dataset (e.g., https://github.com/awesomedata/awesome-public-datasets).

Recall that the Inforestudante platform is limited to 500MB of space, so in cases of large datasets **do not use** *Inforestudante*, instead use external file-sharing services (e.g. Dropbox, Wetransfer). Include the link in your report.

Code

You should use Plotly/Dash and Python to implement the core functionalities of the project. Students are free to use any additional APIs and frameworks as supportive tools (e.g., to compute graph layouts), but teachers are reserved to answer the questions regarding Plotly/Dash.

You should also submit all the code (for visualisation and data preparation) to the *Inforestudante* platform.

Project activities (suggestion)

As a suggestion, this section lists several core activities that students may perform to accomplish the project. This includes the following:

- Definition of target users and a problem that the visualisation should solve;
- Developing the topic of analysis;
- Definition of data and tasks, as well as the specification of the requirements;
- Acquisition of the necessary data and its preparation;
- Identification of an appropriate visualisation method for each of the views of the visualisation application (e.g., a time-series representation, a map-based visualisation);
- Implementation of the chosen views;
- Refinement and definition of visual encodings;
- Implementation of the data transformation functions;
- Refinement of the visualisations;
- Addition of supportive graphical elements, such as titles, captions, legends, etc.
- Implementation of the basic interaction, such as element selection/hovering, range selection, etc.;
- Validation of the employed models using the chosen method (e.g., critical discussion, questionnaire, user study).

Plagiarism and fraud policy

According to the Article 28.º of the disciplinar reglament for the UC students (https://www.uc.pt/regulamentos/ga/vigentes/Regulamento_Disciplinar_Estudantes_UC)

"2 — A fraude cometida em sede de avaliação de uma unidade curricular implica a anulação imediata dessa avaliação e leva à reprovação liminar do estudante na inscrição na unidade curricular em causa, devendo ser registada na plataforma informática da UC e averbada no processo individual do estudante."